INCH-POUND

MIL-S-24187E(SH)
22 March 1991

SUPERSEDING
MIL-S-24187D(SH)
15 June 1978

MILITARY SPECIFICATION

SWITCH ASSEMBLIES, LINEAR MOVEMENT, MANUAL AND REMOTE OPERATION, GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements, operating conditions, and tests for linear movement, multipole, multiposition switch assemblies for manual and remote operation. Switch assemblies are for use in low level circuits and for power circuits and for power circuits up to a maximum of 10 amperes (see 6.1).

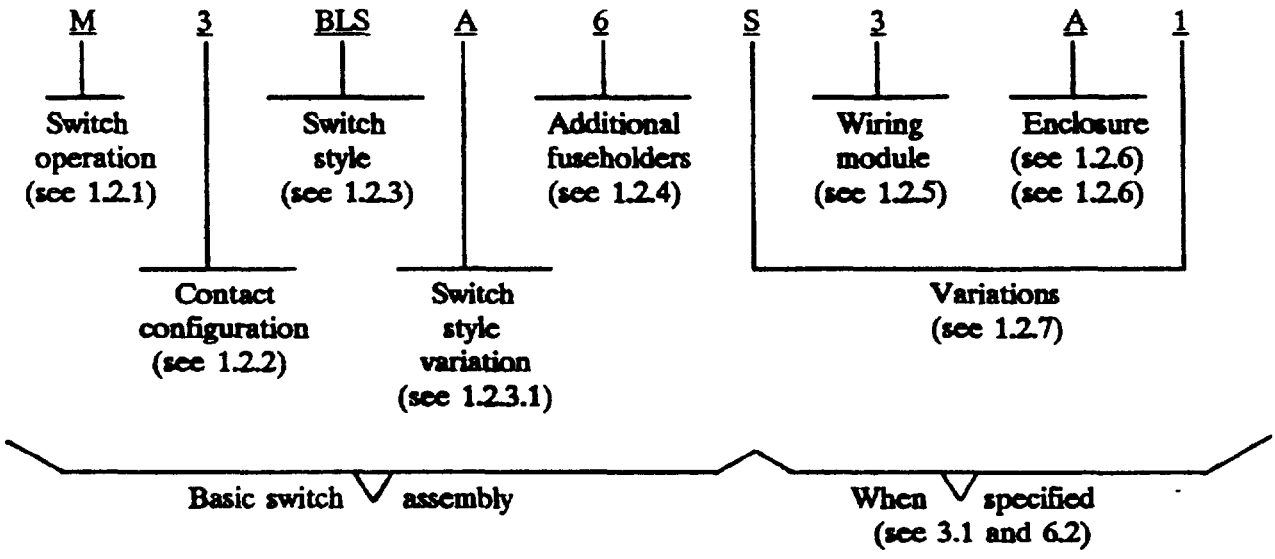
1.2 Classification. The type designation shall be of the following form, and as specified (see 3.1 and 6.2):

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

AMSC N/A FSC 5930

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1.2.1 Switch operation. Switch operation shall be designated by a single letter as follows:

M Manual operation only
R Remote operation with manual override.

1.2.2 Contact configuration. Configuration of contacts on the movable contact plate shall be designated by a single digit as follows and shall be as shown on figure 1:

Configuration 2	Switch action: 1, 2, off or 1, off 2
Configuration 3	Switch action: 1, 2, 3, –N, off ("N" is the number of active positions)
Configuration 4	Switch action: off, 1, 1 and 2, 2
Configuration 4-A	Switch action: 1, off, 2, 1 and 2, 1
Configuration 5	Switch action: 1, 2, 3, 4, 5, off.

1.2.2.1 Contact configuration variation. A letter following the basic switch style shall be used to show a contact variation for that basic switch (see 3.1).

1.2.3 Switch style. Switch style shall be designated by the letters LS, BLS, CLS, and so forth, indicating the basic switch style (see 3.1).

1.2.3.1 Switch style variation. A letter following the basic switch style shall be used to show a switch variation for that basic switch (see 3.1).

1.2.4 Additional fuseholders. Number of additional 2-circuit fuseholders required for the protection of connected load circuits shall be designated by the number specified in the applicable specification sheet (see 6.2) When no additional fuseholders are required the number 0 shall be

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used. Fuseholders required in control circuits of remote operated switch assemblies specified in the applicable specification sheet shall not be designated as additional fuseholders. When the switch style has no provisions for mounting fuseholders, this designator shall be omitted from the type designated.

1.2.5 Wiring module. When required, wiring modules for enclosed switches shall be as specified in the applicable specification sheet and figure 2.

1.2.6 Enclosure. When required, enclosures shall conform to the requirements specified in the applicable specification sheet.

1.2.7 Variations. Variations such as connectors, different circuitry and wiring configurations shall be indicated by the designations "S" and a number modifier (see 3.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-N-290	Nickel Plating (Electrodeposited)
PPP-F-320	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes

MILITARY

MIL-I-631	Insulation, Electrical, Synthetic-Resin Composition, Nonrigid
MIL-S-901	Shock Tests, HI (High-Impact) Shipboard Machinery, Equipment and Systems, Requirements for
MIL-L-3661	Lampholders, Indicator Lights, Indicator-Light Housings, and Indicator Light Lenses, General Specification for
MIL-L-3661/6	Lampholders, Indicator-Lights, Indicator-Light Housings and Indicator-Light Lenses Dusttight Housing, Style LH74

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MIL-L-3661/13	Lampholders, Indicator Lights, Indicator-Light Housings, and Indicator Light Lenses, Dusttight Lenses, Style LC13
MIL-B-5423	Boots, Dust and Water Seal (For Toggle and Pushbutton Switches, Circuit Breakers, and Rotary-Actuated Parts), General Specification for
MIL-T-7928	Terminals, Lug: Splices, Conductor: Crimp Style, Copper, General Specification for
MIL-S-13282	Silver and Silver Alloys
MIL-L-15098	Lamps, Glow, General Specification for
MIL-L-15098/11	Lamps, Glow, Indicator, Types C7A, C9A, A1G and A1H
MIL-F-15160	Fuses, Instrument, Power, and Telephone
MIL-F-15160/77	Fuses, (Indicating), Style F77A
MIL-W-16878	Wire, Electrical, Insulated, General Specification for
MIL-W-16878/1	Wire, Electrical, Polyvinyl Chloride (PVC) Insulated, 105 °C, 600 Volts
MIL-W-16878/4	Wire, Electrical, Polytetrafluoroethylene (PTFE) Insulated, 200 °C, 600 Volts, Extruded Insulation
MIL-E-17555	Electronic and Electrical Equipment Accessories, and Provisioned Items (Repair Parts); Packaging of
MIL-L-19140	Lumber and Plywood, Fire Retardant Treated
MIL-F-19207	Fuseholders, Extractor Post Type, Blown Fuse Indicating and Nonindicating General Specification for
MIL-F-19207/38	Fuseholders, Extractor Post Type, Blown Fuse Indicating, Type FHL57G
MIL-S-23190	Straps, Clamps, and Mounting Hardware, Plastic and Metal for Cable Harness Tying and Support
MIL-G-23827	Grease, Aircraft and Instrument, Gear and Actuator Screw
MIL-C-28731	Connectors, Electrical, Rectangular, Removable Contact, Formed Blade, Fork Type (for Rack and Panel and Other Applications), General Specification for
MIL-C-28731/20	Connector, Electrical, Contact, Fork Type, Crimp, Removable (8.5 Amperes)
MIL-C-28731/23	Connector, Electrical, Contact, Fork Type, Removable, Double Wire Crimp

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MIL-C-28731/35	Connector, Electrical, Contact, Fork Type, Crimp Removable (5.0 Amperes)
MIL-G-45204	Gold Plating, Electrodeposited
MIL-I-45208	Inspection System Requirements

STANDARDS

MILITARY

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-167-1	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-1285	Marking of Electrical and Electronic Parts
MIL-STD-1303	Painting of Naval Ordnance Equipment
MS25100	Switch, Toggle, Two Pole, Sealed Toggle

See supplement 1 for list of associated specification sheets.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual switch assembly requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Switch assemblies furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time of award of contract (see 4.4 and 6.4).

3.3 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable

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without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4 Design and construction (see 6.3 and appendix).

3.4.1 Switch assembly. A switch assembly shall be a grouping of modules which contains all mechanisms and circuitry necessary to perform the required switching functions. The switch assembly shall be designed for panel mounting.

3.4.2 Modular construction. Switch assemblies shall be of modular construction. Modules shall be as specified in the applicable specification sheet. Typical modules are listed below:

a. Front module (see 6.5.1)
b. Mechanism module (see 6.5.2)
c. Remote control module (see 6.5.3)
d. Control circuit module (see 6.5.4)
e. Switch contact module (see 6.5.5)
f. Wiring module (see 6.5.6)
g. Enclosure (see 6.5.7).

3.4.3 Linear movement. In performing the switching function, the moving contact plate of the switch contact module shall move with a linear motion, as contrasted to the rotary motion of the operating handle.

3.4.4 Plug-in connection. Stationary contacts of the switch contact module shall, in addition to engaging the moving switch contacts, engage with the contacts of the switch contact module connector plug.

3.4.5 Enclosure. Enclosure, when required, shall be as specified in the applicable specification sheet.

3.4.6 Materials. Materials shall be as specified herein and in the drawings referenced in the applicable specification sheet.

3.5 Detail requirements.

3.5.1 Toggle switches. Toggle switches used in remote operated switch assemblies shall be in accordance with MS25100, part number MS25100-23. Where moisture seal is required, boots in accordance with MIL-B-5423 shall be used (see 6.2)

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3.5.2 Fuses and fuseholders.

3.5.2.1 Fuses. Fuses shall be in accordance with MIL-F-15160 and MIL-F-15160/77. Unless otherwise specified (see 6.2), fuses shall be provided and installed only in the motor control circuit of remote operated switch assemblies. Fuses shall be provided and installed in the additional fuseholders for the load circuits as specified (see 1.2.4, 3.5.2.2 and 6.2). Fuses having a current rating of less than one ampere shall not be used. Ampere rating of each fuse furnished in a switch assembly shall be marked on the front description plate, adjacent to the fuseholder in which the fuse is contained.

3.5.2.2 Fuseholders. Fuseholders shall be in accordance with MIL-F-19207 and MIL-F-19207/38, type FHL57G. Fuseholders, as required in the applicable specification sheet, shall be provided and connected into the control circuit of each remote operated switch assembly. Additional fuseholders for load circuits shall be provided as specified (see 1.2.4 and 6.2). Blown fuse indicating circuits of fuseholders in a switch assembly shall be connected in parallel and one pair of leads shall be brought out to specified contacts on the assembly as specified in the applicable specification sheet.

3.5.3 Handle. For other than style SLS switches, the position of the switch handle shall be adjustable in 45 degree increments with respect to the flat on the handle shaft. The handle shall contain an indexing washer which may be secured in any of eight positions concentric with the shaft engaging hole. When keyed with the flat on the handle shaft, the indexing washer shall limit the handle to one selected position. For style SLS switches, the switch position pointer assembly in the handle shall be adjustable in 90 degree increments with respect to the flat on the handle shaft.

3.5.4 Indicator lights. Indicator lights required for remote operated switch assemblies shall be in accordance with MIL-L-3661, and as specified in the applicable specification sheet. Lamp housings shall be type LH74 in accordance with MIL-L-3661/6. Lens shall conform to MIL-L-3661/13, style LC13. Neon lamps shall be in accordance with MIL-L-15098 and MIL-L-15098/11 (part number M15098/11-001, Industry number C7a).

3.5.4.1 Color designation. Lens color for the indicator lights shall be as follows:

a. Nonsynchronous light – yellow (LC13YN2)
b. Motor overload light – red (LC13RN2).

3.5.5 Connectors. Connectors shall be as specified in the applicable specification sheet.

3.5.5.1 Control circuit module connector. Connector for the control circuit module (internal to the basic switch assembly) shall be as specified in the applicable specification sheet.

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3.5.5.2 Switch contact module connector plug. Switch contact module connector plug(s) shall be in accordance with the applicable specification sheet. Plug(s) shall be furnished as part of the wiring modules in enclosed switches. Unless otherwise specified (see 6.2), plugs shall not be furnished as parts of non-enclosed switches.

3.5.5.3 Other connectors. Other connectors shall be in accordance with MIL-C-28731 or as specified in the applicable specification sheet. Both plug and receptacle shall be provided for connectors integral to the switch, unless otherwise specified in the applicable specification sheet.

3.5.5.4 Ship's wiring connectors.

3.5.5.4.1 Ship's wiring connector receptacles. Ship's wiring connector receptacles shall be in accordance with MIL-C-28731 or as specified in the applicable specification sheet. Receptacles shall be furnished as parts of the wiring modules in enclosed switches and they shall be mounted at the rear of the switch.

3.5.5.4.2 Ship's wiring connector plugs. Ship's wiring connector plugs shall be in accordance with MIL-C-28731 or as specified in the applicable specification sheet. They shall be capable of mating with the ship's wiring connector receptacles. Unless otherwise specified (see 6.2), ship's wiring connector plugs shall not be supplied with the switch assembly.

3.5.5.5 Connector contacts. Single wire contacts applicable to MIL-C-28731 connectors only, shall be in accordance with MIL-C-28731, MIL-C-28731/20, or MIL-C-28731/35. Double wire contacts shall be in accordance with MIL-C-28731/23. Wires or combinations of wires that may be used are shown in table I.

TABLE I. *Crimp contact wire sizes.*

	Wire sizes	
MIL-C-28731/20	MIL-C-28731/23	MIL-C-28731/35
No. 18 to No. 26	Two No. 16 Two No. 18 Two No. 20 One No. 16 and one No. 20	No. 22 to No. 30

3.5.6 Wiring.

3.5.6.1 Internal wire. Internal wire shall be in accordance with MIL-W-16878 and MIL-W-16878/1, type B, with a polyamide jacket and shall have a minimum of 19 strands. The polyamide jacket may be omitted from internal wiring. An alternate wire shall be in accordance with MIL-W-16878 and MIL-W-16878/4, type E with silver coat and teflon, or equal, insulation

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3.5.6.2 Protection of wiring. Edges of parts that come into contact with wiring shall be rounded. Metal parts shall be provided with rounded insulating guards, grommets, or pads to prevent damage to the wiring. Stand-offs and protective plates shall be provided where required to prevent wiring from being damaged by terminal nuts or screw threads. Insulating guards, grommets, pads, stand-offs and protective plates shall be type F, grade A in accordance with MIL-I-631. Nylon straps used for bundling wires shall be in accordance with MIL-S-23190; lacing twine, if used, shall be of suitable size. Care should be exercised in tying and lacing to assure that wiring bundles are secure, but not so tight as to cut the primary insulation during normal service.

3.5.6.3 Wire termination. Terminal lugs in accordance with MIL-T-7928, type II, shall be used to terminate wires to screw or stud terminals.

3.5.6.4 Splices. In-line wiring splices shall not be used.

3.5.7 Switching contacts. Stationary and movable switching contacts shall be in accordance with the details shown on the drawings specified in the applicable specification sheet and in 3.5.7.1 and 3.5.7.2.

3.5.7.1 Movable contacts. Movable contacts shall be beryllium copper with a coin silver interface in accordance with MIL-S-13282, grade C on the contact surface.

3.5.7.2 Stationary contacts. Stationary contacts shall be phosphor bronze (spring temper) with a fine silver interface in accordance with MIL-S-13282, grade A on the switching side. Contacts shall have a minimum of 0.000050 inch gold plating in accordance with MIL-G-45204, type II, grade D. When specifically approved, a selective plating process may be used so that only the contact engagement area has the silver interface specified and only the contact termination area has the 0.000050 inch gold plating over 0.000050 nickel (minimum) in accordance with QQ-N-290. The remaining portion of the contact shall have the nickel finish, but need not have any overplating.

3.5.8 Lubrication.

3.5.8.1 Mechanical parts. Moving mechanical parts shall be lubricated with a compound in accordance with MIL-G-23827.

3.5.8.2 Switch contacts. A minute amount of lubricant conforming to MIL-G-23827 shall be applied to the switch contacts.

3.6 Electrical requirements.

3.6.1 Electrical rating. Electrical rating of the switch assembly shall be as specified in the applicable specification sheet (see 3.1).

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3.6.2 Internal wiring. Internal wiring of switch assemblies (for fuseholders, wiring modules, and remote control circuitry) shall be as specified in the applicable specification sheet (see 4.8.1.1).

3.6.3 Remote control. Remote control circuit for remote operated switch assemblies shall be as specified in the applicable specification sheet. Remote control circuit shall cause a drive motor to move the switch contacts to the ordered position. After the movable contacts have reached an ordered position, maintenance or removal of the control signal shall not affect the position of the contacts.

3.6.4 Contact configurations. Switching contact groups or poles in a switch assembly shall have the same configuration (2, 3, or 4) as specified in 1.2.2 and 4.8.1.3 and as specified in the applicable specification sheet. Contacts used for switch control and position indicating circuits are exempt from this requirement.

3.6.5 Electrical operation.

3.6.5.1 Load circuit operation. Switch assembly shall make and break the required load circuits in all positions and in all contact groups (see 4.8.1.3).

3.6.5.2 Simultaneous operation. Throughout the switch assembly, "break-before-make" contacts shall open before any contact closes in the adjacent position. "Make-before-break" contacts shall close in the adjacent position before opening of any circuit of the original position (see 4.8.1.4.).

3.6.5.3 Actuating time (remote operated switch assemblies only). The actuating times for a single position and a multiposition change shall not exceed the values specified in the applicable specification sheet (see 4.8.2.1).

3.6.5.4 Transfer time (remote operated switch assemblies only). The transfer time shall not exceed the values specified in the applicable specification sheet (see 4.8.2.2).

3.6.5.5 Current drain (remote operated switch assemblies only). The maximum current drain per phase for any condition of operation (starting, operating or motor stalled) shall be as specified in the applicable specification sheet (see 4.8.2.3).

3.6.6 Ambient temperature. Switch assemblies shall operate in any ambient temperature within the limits specified in the applicable specification sheet (see 4.8.10.4).

3.6.7 Endurance. Switch assembly shall make, break, and carry its rated currents and voltages for the number of operations specified in the applicable specification sheet. Switch assembly shall remain electrically and mechanically operative during and following the endurance specified (see 4.8.10). There shall be no evidence of intermittent operation during or following the endurance test.

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3.6.8 Contact resistance. Contact resistance shall not exceed the values specified in the applicable specification sheet (see 4.8.8).

3.6.9 Total circuit resistance. Total circuit resistance shall not exceed the values specified in the applicable specification sheet (see 4.8.9).

3.6.10 Dielectric withstanding voltage. Switch assembly shall withstand 1200 volts root mean square (rms) without flashover or breakdown (see 4.8.6). Following the endurance test, switch shall withstand 600 volts rms.

3.6.11 Insulation resistance. The insulation resistance shall be not less than 10 megohm (see 4.8.7).

3.6.12 Temperature rise. Temperature rise at the switch stationary contacts shall not exceed the limits specified in the applicable specification sheet (see 4.8.11).

3.7 Mechanical requirements.

3.7.1 Vibration. There shall be no evidence of electrical or mechanical damage (see 4.8.12). There shall be no contact transfer and momentary opening of closed contacts shall not exceed 10 microseconds.

3.7.2 Shock. The switch assembly shall withstand the shock test without evidence of electrical or mechanical damage (see 4.8.13). There shall be no contact transfer and momentary opening of closed contacts shall not exceed 0.015 second.

3.7.3 Torque. The torque required at the handle shaft to operate the switch assembly shall not exceed the values specified in the applicable specification sheet (see 4.8.4).

3.7.4 Stop strength. Stops shall withstand the specified torque as shown in the applicable specification sheet (see 4.8.5).

3.7.5 Humidity. Switch assembly insulation resistance shall be not less than 1 megohm while wet nor less than 10 megohm after drying (see 4.8.14). At the conclusion of the test, there shall be no evidence of corrosion, breaking, cracking, or spalling that will affect the operation of the switch assembly.

3.7.6 Contact and bushing retention.

3.7.6.1 Stationary contacts - switch contact module. Stationary contacts shall withstand an axial load of 5 pounds. There shall be no displacement of the bushings or contacts and there shall be no damage to the bushings, contacts, or the stationary plate (see 4.8.3.1).

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3.7.6.2 Stationary contact bushing. Stationary contact bushings shall withstand an axial load of 15 pounds. There shall be no displacement of the bushings or the contacts and there shall be no damage to the bushings, contacts, or the stationary contact plate (see 4.8.3.1).

3.7.6.3 Connector contact retention. Individual contacts shall withstand an axial load of 15 pounds without damage to the contact, insulator, or contact retaining clip (see 4.8.3.2).

3.7.7 Switch contact module connector engagement. There shall be no damage to contacts in the plug or receptacle, no displacement of contacts, no damage to or displacement of alignment devices and securing devices of the switch contact module connector (see 4.8.10.5).

3.8 Interchangeability. Modules (see 3.4.2) shall be electrically and mechanically interchangeable with corresponding modules of the same switch style regardless of the manufacturer (see 4.8.15).

3.9 Marking.

3.9.1 Switch assembly and module identification. Switch assemblies and modules shall be legibly marked in accordance with MIL-STD-130 with the following information:

a. Manufacturer's name, trademark, or code in accordance with MIL-STD-1285

b. Switch type designation (see 1.2), if applicable

c. Date code showing the year and week of manufacture in accordance with MIL-STD-1285

d. Other marking as shown on the drawings specified in the applicable specification sheet.

3.9.2 Front description plate. Front description plate (see 6.5.8) markings shall be in accordance with the drawings specified in the applicable specification sheet.

3.9.3 Terminal identification. Terminals or contact positions shall be identified as shown in the applicable specification sheet.

3.10 Onboard repair parts. When required (see 6.2), onboard repair parts shall be furnished in accordance with the tables in the applicable specification sheet in the quantities as specified (see 6.2).

3.11 Painting. Unless otherwise specified (see 6.2), paint shall be light gray enamel in accordance with system No. 37 of MIL-STD-1303. For submarine applications, shall be painted white enamel in accordance with No. 13 of MIL-STD-1303.

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3.12 Workmanship. Switch assemblies and modules shall be processed in such a manner as to be free of cracked or displaced parts, sharp edges, burrs, and other defects which will affect the life, serviceability, or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208 (see 6.2).

4.1.2 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program (see 6.2 and 6.3). The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed at the ambient temperature and humidity conditions specified in the "General requirements" of MIL-STD-202 and electrical operations, tests, and measurements shall be made at rated voltage, frequency, and loads specified in the applicable specification sheet.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Materials inspection (see 4.3)
b. Qualification inspection (see 4.4)
c. Quality conformance inspection (see 4.6).

4.3 Materials inspection. Materials inspection shall consist of verification that the materials used in fabricating the switch assemblies are in accordance with the applicable referenced specifications or requirements prior to such fabrication (see 6.3)

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4.4 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to the Naval Sea Systems Command (NAVSEA), on sample switch assemblies produced with equipment and procedures normally used in production. Qualification inspection shall consist of the examination and tests listed in table II, in the order shown.

4.4.1 Samples for qualification. Two sample switch assemblies of each contact configuration specified in the individual specification sheets (see 3.1) shall be subjected to the examination and tests specified in table II. If a specification sheet covers both manual and remote operated switch assemblies, the remote operated switch assemblies shall be submitted for qualification testing. Qualification of the remote operated switch assemblies shall be submitted for qualification testing. Qualification of the remote operated switch assemblies shall include qualification of the manual operated switch assemblies of the same contact configuration. The samples shall be designated "A" and "B". Failure of any qualification sample to comply with examinations and tests listed in table II shall result in rejection of the qualification sample.

4.5 Sampling for quality conformance inspection.

4.5.1 Inspection lot. All switch assemblies and modules (repair parts) presented for delivery at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period.

4.5.2 Group A examination and tests. All switch assemblies and modules (repair parts) in each inspection lot shall be subjected to the group A examination and tests listed in table III. The results of each test shall be compared with specification requirements. Failure to conform to this specification for any group A examination or test shall be counted as a defect, and the switch assemblies and modules shall not be offered for delivery until the defects have been corrected and the switch assemblies and modules reexamined and retested.

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TABLE II. *Qualification inspection.*

Inspection	Sample letter and test sequence number		Requirement	Test method
	A	B		
General examination	1	1	3.4, 3.5	4.7
Material inspection	1	1	3.4.6	4.3
Electrical operation:				
Internal wiring	2	2	3.6.2	4.8.1.1
Remote control	2	2	3.6.3	4.8.1.2
Contact configuration	2	2	3.6.4	4.8.1.3
Load circuit operation	2	2	3.6.5.1	4.8.1.3
Simultaneous operation	2	2	3.6.5.2	4.8.1.4
Performance.				
Actuating time	3	3	3.6.5.3	4.8.2.1
Transfer time	3	3	3.6.5.4	4.8.2.2
Current drain	3	3	3.6.5.5	4.8.2.3
Contact and bushing retention	4	–	3.7.6	4.8.3
Torque (room temperature)	5	4	3.7.3	4.8.4
Stop strength	6	5	3.7.4	4.8.5
Dielectric withstanding voltage	7	6	3.6.10	4.8.6
Insulation resistance	8	7	3.6.11	4.8.7
Contact resistance	9	8	3.6.8	4.8.8
Total circuit resistance	10	9	3.6.9	4.8.9
Endurance:				
Temperature rise	11	10	3.6.12	4.8.11
Mechanical operation	12	11	3.6.7	4.8.10.1
Electrical load	13	12	3.6.7, 3.6.5.1	4.8.10.3
Ambient temperature	14	13	3.6.6	4.8.10.4
Engagement of connector plug	15	–	3.7.7	4.8.10.5
Contact resistance	16	14	3.6.8	4.8.8
Vibration	17	–	3.7.1	4.8.12
Shock	18	–	3.7.2	4.8.13
Torque (room temperature)	19	15	3.7.3	4.8.4
Stop strength	20	16	3.7.4	4.8.5
Humidity	–	17	3.7.5	4.8.14
Dielectric withstanding voltage	21	18	3.6.10	4.8.6
Insulation resistance	22	19	3.6.11	4.8.7
Simultaneous operation	23	20	3.6.5.2	4.8.1.4
Actuating time	24	21	3.6.5.3	4.8.2.1
General examination	25	22	3.4, 3.5	4.7
Module interchangeability	26	–	3.8	4.8.15

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TABLE III. *Quality conformance inspection.*

Inspection	Requirement	Test method
Group A		
Switch assemblies		
General examination	3.4, 3.5	4.7
Electrical operation:		
Internal wiring	3.6.2	4.8.1.1
Remote control	3.6.3	4.8.1.2
Contact configuration	3.6.4	4.8.1.3
Load circuit operation	3.6.5.1	4.8.1.3
Simultaneous operation	3.6.5.2	4.8.1.4
Contact resistance	3.6.8	4.8.8
Total circuit resistance	3.6.9	4.8.9
Dielectric withstanding voltage	3.6.10	4.8.6
Torque		3.7.34.8.4
Stop strength	3.7.4	4.8.5
Module interchangeability	3.8	4.8.15
Modules (repair parts)		
General examination	3.4, 3.5	4.7
Module certification	–	4.8.15.1
Group B		
Performance.		
Actuating time	3.6.5.3	4.8.2.1
Transfer time	3.6.5.4	4.8.2.2
Current drain	3.6.5.5	4.8.2.3
Insulation resistance	3.6.11	4.8.7
Contact and bushing retention[1]	3.7.6	4.8.3
Group C		
All of the examination and tests listed in table II		

[1]Contact retention may be conducted as part of in-process manufacturing procedures.

4.5.3 Sampling for group B tests. A sample of switch assemblies which have met the group A examination and tests shall be selected from each inspection lot in accordance with table III and subjected to the group B tests listed in table IV. The results of each test shall be compared with specification requirements If one or more defects are found in any sample, the entire lot shall be rejected and screened by 100 percent by the contractor for the defect found

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TABLE IV. *Sampling for group B tests.*

Lot size	Critical characteristic	Major characteristic	Minor characteristic
2 to 8	All	All	3
9 to 15	All	All	3
16 to 25	All	20	3
26 to 50	All	20	5
51 to 90	All	20	6
91 to 150	125	20	7
151 to 280	125	20	10
281 to 500	125	47	11
501 to 1200	125	47	15
1201 to 3200	125	53	18
3201 to 10,000	192	68	22
10,001 to 35,000	294	77	29
35,001 to 150,000	294	96	29

4.5.4 Sampling for group C examination and tests. Group C examination and tests shall be conducted at least once every 5 years on one switch of each type (see 3.1). The switches shall be subjected to the tests shown for sample "A" in table II.

4.6 Quality conformance inspection. Switch assemblies selected in accordance with 4.5 shall be subjected to the examination and tests listed in table III in the order shown. Each examination and test is classified as critical.

4.7 General examination. Switch assembly and modules (repair parts) shall be examined for conformance to referenced specifications and drawings with respect to safety, material, finish, workmanship, construction, assembly, dimensions, marking (including module and designation plate marking), fuseholder quantity and location. Inspection shall be limited to those tests which will not require disassembling switches or modules except for removal of switch enclosures where provided. In-process inspection shall be used to determine conformance of each module to critical dimensional interchangeability and performance requirements (see 3.8 and 4.8.15). For group A inspection, "go-no-go" types of inspection fixtures or gauges may be used in lieu of dimensional examination to determine compliance with the dimensional requirements.

4.8 Test methods. Where remote operation of a switch assembly is required, a power supply and test circuit equivalent to figures 3 or 4 (as applicable), shall be provided. Circuit connections to switch assemblies for the various tests shall be made through separate switch contact module connector plugs or ship's wiring connector plugs, as applicable. Tests requiring electrical measurements shall be performed using the appropriate connector plugs (see figure 2).

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4.8.1 Electrical operation.

4.8.1.1 Internal wiring. Switch assembly shall be examined to verify that the internal wiring meets the requirements of 3.6.2.

4.8.1.2 Remote control. For remote operated switch assemblies, a test shall be conducted to verify that the switch assembly responds to a specified control signal as shown in the applicable specification sheet for both a one position and a maximum position movement of the switch in both directions. For switch assemblies having remote/manual toggle switches in their control circuits, the toggle switches shall be operated to verify that they operate both mechanically and electrically to control the remote/manual feature of the switches.

4.8.1.3 Contact configuration and load circuit operation. Load contacts of the switch assembly shall be wired to demonstrate compliance with the required switching configuration (see 3.6.4). Lamps or other indicating means shall be used to verify that the configuration is correct. The test shall also be used to verify proper location of the stops (when applicable). Switch shall be operated manually for this test.

4.8.1.4 Simultaneous operation. Contacts in the same relative position in the horizontal (lettered) rows of the contacts shall be connected in parallel and wired through a neon indicator light for each group. In switching from one position to the next, all lights, except for those connected to "make-before-break" contacts in switch configuration 4 (see figure 1), shall extinguish momentarily. Switching shall be manual and sufficiently slow to observe extinguishing of the lamps.

4.8.2 Performance (remote operated switch assemblies).

4.8.2.1 Actuating time. Actuating time for an individual pole shall start with the application of the control voltage and end when contact is made in the commanded position. Command for a multiposition change shall be applied directly without sweeping through the commands for intermediate positions. Actuating times shall be recorded for each actuation and shall meet the requirements of 3.6.5.3. The following measurements shall be made in room ambient conditions:

a. Time required for single position changes in both directions of travel for each pole being tested.

b. Time required for a multiposition change of the specified number of positions (see applicable specification sheet) in both directions of travel for each pole being tested.

c. Measurements shall be made on randomly selected poles. For group B tests, 4 poles shall be measured for switch assemblies having 60 poles or less; 6 poles shall be measured for switch assemblies having 61 – 100 poles; and 8 poles shall be measured for switch assemblies having more than 100 poles. For qualification and group C tests, measurements shall be made on 50 percent of the poles of the switch assembly.

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4.8.2.2 Transfer time. Transfer time for an individual pole shall be measured from the time the initially closed contact opens to the time the initially open contact closes for a one position change. Measurements shall be made for single position changes in both directions of travel for each pole being tested. The test shall be performed in room ambient conditions. Number of poles tested shall conform to 4.8.2.1(c). The transfer time shall be recorded for each actuation of each pole being tested and shall meet the requirements of 3.6.5.4.

4.8.2.3 Current drain. With rated voltage applied, the current in each line of the supply to the switch drive motor shall be monitored in the three conditions specified (see 3.6.5.5). Line current shall not exceed the value specified in 3.6.5.5.

4.8.3 Contact and bushing retention.

4.8.3.1 Stationary contacts and bushings. Specified loads (see 3.7.6.1 and 3.7.6.2) shall be applied externally to contacts and bushings mounted in a stationary contact plate. Applicable loads shall be applied to the split ends of the contacts and to the round ends of the bushings on the connector side of the contact plate. The two loads need not be applied simultaneously. A reduced load shall be applied to the part being tested and, after the slack between the part and its mounting has been taken up, the load shall be increased uniformly at 1 pound per second until the specified load is reached. The specified load shall be maintained for 5 seconds. During and after application of the specified load, the contacts and bushings shall comply with the requirements of 3.7.6.1 and 3.7.6.2. For qualification and group C tests, a minimum of 20 contacts and bushings evenly spaced over the switch contact plate shall be tested. For group B tests, conducted when switches are offered for quality conformance inspection, 5 contacts and bushings spaced around the periphery of the switch contact plate shall be tested. For qualification or group B tests, the contact and bushing retention test may be made on completed contact plates prior to their mounting in switch assemblies.

4.8.3.2 Switch contact module connector plug contacts. Specified maximum axial tension load shall be applied to each of 20 contacts spaced over the connector. Load shall be applied uniformly at the rate of 1 pound per second through a size 16 AWG wire crimped to each contact. Contacts shall meet the requirements of 3.7.6.3 after three insertions and withdrawals from the same hole. This test is not required of connectors conforming to MIL-C-28731.

4.8.4 Torque (rotational). Maximum torque required at the handle shaft to operate the switch shaft with respect to the switch body, shall be measured by a torque wrench or other torque measuring device. Torque shall not exceed the requirements specified in 3.7.6.3. Switch shall be tested in each position and in both directions. For qualification and group C testing, the tests shall be conducted immediately after exposure for 2 hours to the maximum temperature and then again immediately after exposure for 2 hours to the minimum temperature specified in the applicable specification sheet. For qualification and group C testing, record the highest torque between all switch positions for five complete rotations. For group A testing, record the highest torque between all switch positions for one complete clockwise and one complete counterclockwise rotation

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4.8.5 Stop strength. With the switch stops in place, the switch shaft shall be rotated to the clockwise and counterclockwise extremes, and the torque specified in 3.7.4 applied to the switch shaft against the stop.

4.8.6 Dielectric withstanding voltage. Switch assembly shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

a. Test voltage shall be 1200 volts, room mean square (rms), 60 hertz (Hz), except that following the endurance tests the test voltage shall be 600 volts, rms, 60 Hz.

b. For group A inspection, the test voltage shall be applied for 5 seconds. For qualification and group C inspection, the test voltage shall be applied for 1 minute.

c. In each switch position, the test voltage shall be applied between open circuit contacts and between closed contacts and noncurrent carrying parts.

d. The test shall apply to stationary and movable control and switching contacts only.

e. For enclosed switches, the test voltage shall be applied at the ship's wiring connector receptacles (see figure 2).

NOTE

Dielectric withstanding voltage shall not be applied to circuitry which includes components which would normally be damaged by the test voltage (that is, control circuit module, transient suppression diode).

4.8.7 Insulation resistance. Switch assembly shall be tested in accordance with method 302 of MIL-STD-202 The following details shall apply:

a. Test condition B shall be used.

b. In each switch position, the test voltage shall be applied between open circuit contacts, and between closed contacts and noncurrent carrying parts.

c. For group B tests, the test voltages shall be applied for 15 seconds. For qualification and group C inspection, the test voltage shall be applied for 1 minute.

NOTE

The insulation resistance test voltage shall not be applied to circuitry which includes components which would normally be damaged by the test voltage (that is, control circuit module, transient suppression diode).

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4.8.8 Contact resistance. Switch assemblies shall be tested in accordance with method 307 of MIL-STD-202. Measurements shall be taken at the crimp portion of the contacts in the switch contact module connector plug (see figure 2). The following details shall apply:

a. Method of connection – Copper leads, size 16 American Wire Gauge (AWG) for switches rated over 4 amperes and size 20 AWG for switches rated at 4 amperes or less, shall be crimped to connector contacts; the contacts inserted into a connector plug and fastened to the switch.

b. Test load shall be as shown on the applicable specification sheet. For group A inspection, any load from 1 ampere to the maximum rated load of the switch may be used at any convenient voltage within the switch rating.

c. Resistance shall be measured before, during (every 10,000 cycles), and after endurance test on the same contacts used for the endurance test, for qualification and group C tests. For group A inspection, 20 percent of the poles (to a maximum of 20 poles) randomly selected from locations throughout the switch shall be tested.

d. Prior to each measurement, the switch shall be actuated one time.

e. For qualification and group C tests, the average value of five separate measurements for each circuit tested shall be determined. For group A inspection, one measurement shall be taken for each circuit being tested.

4.8.9 Total circuit resistance. Total circuit resistance shall be measured at the crimp portion of the contacts in the ship's wiring connector plug (see figure 2). The following details shall apply:

a. Method of connection – Copper leads, size 16 AWG for switches rated over 4 amperes and size 20 AWG for switches rated at 4 amperes or less, shall be crimped to connector contacts; the contacts inserted into a connector plug and fastened to the switch.

b. Test load shall be as shown on the applicable specification sheet. For group A inspection, any load from 1 ampere to the maximum rated load of the switch may be used at any convenient voltage within the switch rating.

c. Resistance shall be measured before, during (every 10,000 cycles), and after endurance test on the same contacts used for the endurance test, for qualification and group C tests. For group A inspection, 20 percent of the poles (to a maximum of 20 poles) randomly selected from locations throughout the switch shall be tested.

d. Prior to each measurement, the switch shall be actuated one time.

e. For qualification and group C tests, the average value of five separate measurements for each circuit tested shall be determined. For group A inspection, one measurement shall be taken for each circuit being tested.

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4.8.10 Endurance. Switch assemblies shall be tested to determine conformance to 3.6.7. For both remote and manually operated switch assemblies, the switch assembly shall be operated by the switch handle shaft for 1,000 operations at room ambient conditions without electrical load. Remote operated switch assemblies shall complete the specified additional operations by means of their remote operating mechanisms. Manual switch assemblies shall be operated by the handle shaft for the additional operations (see applicable specification sheet). Each operation shall consist of the maximum travel of the switch and return to the original position (see applicable specification sheet). Adjustments, replacement of parts, or application of additional lubrication shall not be permitted during the endurance test.

4.8.10.1 Operation. Unless otherwise specified in the applicable specification sheet the rate of operation shall be 10 per minute; however, switch assemblies with contact configuration 3 may be operated at a slower rate if necessary. Mechanism shall come to a stop before application of the appropriate control potential for each reversal in direction of movement. For the type 3 contact configuration switch assemblies, at least 50 percent of the operations in each of the temperature conditions specified shall have the switch assembly mechanism come to a complete stop at each position before application of the appropriate control potential to cause the switch to move to the next position.

4.8.10.2 Positioning of switch assembly under test. For qualification tests, one switch assembly of each type tested shall be mounted in the horizontal position (vertical panel); one switch assembly of each type shall be mounted in a vertical position (horizontal panel). For group C tests, the switch assembly shall be mounted in the horizontal position.

4.8.10.3 Electrical load. Electrical loads shall be applied as specified in the applicable specification sheet.

4.8.10.4 Ambient temperature. Following the 1,000 handle shaft operations (see 4.8.10), the next 5,000 operations of the endurance test shall be performed at the high ambient temperature and humidity specified in the applicable specification sheet. The following 5,000 operations shall be performed at the low ambient temperature specified in the applicable specification sheet. The remainder of the test shall be conducted at the ambient temperature and humidity specified in 4.1.2. Switch assemblies shall be subjected to the high and low ambient temperatures for 2 hours prior to beginning the 5,000 operations. During the high temperature test, the motor thermostats shall be short circuited.

4.8.10.5 Engagement of connector plug. Switch contact module connector plug (see figure 2) shall be disengaged and re-engaged 100 times to the same switch section to determine compliance with 3.7.7.

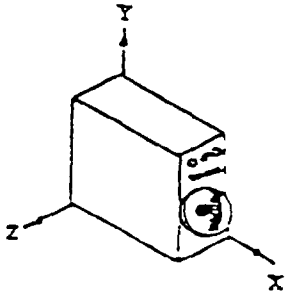
4.8.11 Temperature rise. Temperature rise test shall be made with the switch assembly at room temperature. Contacts used for the endurance test shall be continuously energized at the current specified for the endurance test at any convenient voltage within the rating of the switch Contacts shall be loaded for 4 hours Thereafter, measurements shall be taken at half-hour

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intervals until three successive measurements indicate that the temperature of the energized contacts has been stabilized within 1.8 degrees Fahrenheit (°F) (1 degree Celsius (°C)). Measurements shall be made by means of thermocouples of small wire gauge (28–32 AWG). Temperature sensitive elements shall be placed on the stationary contacts where they first emerge from the switch contact module (see figure 2). Temperature rise shall not exceed the values specified in 3.6.12.

4.8.12 Vibration. Switch assembly shall be tested in accordance with MIL-STD-167-1, type I (environmental vibration) except that the variable frequency test shall be omitted. All groups of closed switching contacts shall be series connected. Circuit shall be monitored in accordance with method 310 of MIL-STD-202 to determine compliance with 3.7.1. Remote operated switch assembly shall be remotely actuated through one full cycle each hour during the test. After test, the switch assembly shall be electrically and mechanically operable. There shall be no evidence of damage of any kind.

4.8.13 Shock. Switch assembly shall be subjected to grade A, class I, type C shock tests in accordance with MIL-S-901 (see 3.7.2). Closed circuit contacts shall be connected in series and shall be monitored for opening at each blow in accordance with method 310 of MIL-STD-202 to determine compliance with 3.7.2. Prior to each hammer drop, the switch mechanism shall be operated to the position specified below.



	Hammer drop		
Axis	1	2	3
X	CCW	C	CW
Y	C	CW	CCW
Z	CW	CCW	C

C = Approximate center of handle rotation
CW = Extreme clockwise handle rotation
CCW = Extreme counterclockwise handle rotation

4.8.14 Humidity. Switch assembly shall be tested in accordance with method 103 of MIL-STD-202. The following details shall apply:

a. Test condition B

b. Switch assembly shall be connected and the voltages applied as specified in the endurance test (see 4.8.10.3). Switch assembly shall be in the off position for the period of the test.

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c. Insulation resistance (see 4.8.7) shall be measured immediately after the switch assembly is removed from the chamber without wiping or drying, to determine compliance with 3.7.5. Remote operated switch assemblies shall be operated to determine compliance with 4.8.1.2. Manual operated switch assemblies shall be actuated to check for freedom of operation. After a drying period, not to exceed 24 hours, insulation resistance (see 4.8.7) shall again be measured to determine compliance with 3.7.5.

4.8.15 Module interchangeability. Each module (see 3.4.2) shall be checked for compliance with 3.8 during the qualification and group C testing by the use of "go-no-go" gauges. Modules or complete switch assemblies may be submitted for interchangeability tests in gauges as specified in the applicable specification sheet. A set of these gauges is maintained by NAVSEA.

4.8.15.1 Module (repair parts) certification. Module certification shall consist of a statement, signed by a responsible official of the switch manufacturing company, attesting that each module being offered as a repair part on a contract or order has been manufactured with identical materials and examined and tested in accordance with the same quality assurance procedures used to assure acceptability of like modules offered as parts of complete switch assemblies manufactured to this specification.

4.9 Inspection of packaging. Sample packages and packs and the inspection of the preservation, packing, and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging or preparation for delivery requirements of referenced documents listed in section 2, see 6.7.)

5.1 General.

5.1.1 Navy fire-retardant requirements.

a. *Treated lumber and plywood.* Unless otherwise specified (see 6.2), all lumber and plywood including laminated veneer materials used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated materials conforming to MIL-L-19140 as follows:

Levels A and B	Type II – Weather resistant Category I – General use
Level C	Type I – Non-weather resistant Category I – General use

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b. *Fiberboard.* Fiberboard used in the construction of interior (unit and intermediate) and exterior fiberboard boxes including interior packaging forms shall conform to the class-domestic/fire-retardant or class-weather resistant/fire-retardant material requirements as specified (see 6.2), of PPP-F-320.

5.2 Packaging requirements. The packaging (preservation, packing, and marking) requirements shall be in accordance with MIL-E-17555 for the level of preservation (A, B, C, or commercial), the level of packing (A, B, C, or commercial), and marking including bar coding and other packaging acquisitioning options therein as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Switch assemblies are intended for use as components of switchboards. Typical applications are command and control switchboards, fire control switchboards, battle telephone switchboards, action cut-out switchboards, and radar distribution switchboards. Remote operated switch assemblies provide for control of the switchboard from one or more remote stations. Separately-mounted enclosed switch assemblies may be used as single units.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification and the applicable specification sheet

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. Type designation (see 1.2 and 3.1)

d. Number and location of additional fuseholders, if required (see 1.2.4 and 3.5.2.2)

e. Handle for style SLS switches, if required (see 3.1 and 3.5.3)

f. Connector plates for style SLS switches (see 3.1)

g. Wiring module connections, if other than specified (see 3.1)

h. Accessories required (see 3.1)

i. Moisture seal of switch shaft, if required (see 3.5)

j. Moisture seal of toggle switches, if required (see 3.5.1)

k. Ampere rating for fuses, if required (see 3.5.2.1)

l. Whether the switch contact module connector plugs for unenclosed switch assemblies are to be furnished (see 3.5.5.2)

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m. Whether the ship's external mating connector plugs for enclosed switch assemblies are to be furnished (see 3.5.5.4.2)

n. Markings required for the switch description plate (see 3.9.2)

o. Rotary description plate markings, if required

p. Quantity of repair parts required (see 3.10)

q. Painting (see 3.11)

r. Inspection requirements (see 4.1.1)

s. When fire-retardant lumber and plywood is not required (see 5.1.1.a.)

t. Class of fire-retardant fiberboard required (see 5.1.1.b.)

u. Level of preservation, level of packing and other packaging options required (see 5.2)

v. Tools, if required (see 6.6).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.4 and Appendix	DI-MISC-80652	Technical Information Reports	–
4.3	DI-E-2121	Certificate of Compliance	–
4.6	DI-T-2072	Report, Test	–

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time of award of contract, qualified for inclusion in the applicable Qualified Products List QPL-24187 whether or not such products have actually been so listed by the date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is

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the Naval Sea Systems Command, Department of the Navy, Washington, DC 20362-5101, and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.4.1).

(Copies of "Provisions Governing Qualification SD-6" may be obtained from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

6.5 Definitions.

6.5.1 Front module. Front module consists of the casting or front plate, description plate and, when required, the fuseholders, indicator lights, and remote-manual toggle switch and associated hardware.

6.5.2 Mechanism module. Mechanism module includes the gearing and associated shafts, including the switch operating shaft, casting, and associated hardware required for manual operation of the switch.

6.5.3 Remote control module. Remote control module includes the motor, motor protection devices, control transformers, the internal wiring connector, fuseholder connector receptacle, frame, and associated hardware.

6.5.4 Control circuit module. Control circuit module includes the printed circuit board, the control relays, components such as resistors and associated mounting hardware.

6.5.5 Switch contact module. Switch contact module includes the moving and stationary switch contacts, switch contact plates, housing, and associated hardware.

6.5.6 Wiring module. Wiring module is composed of the connectors, connector contacts, wire, and associated hardware required to interface between the switch contact module and the ship's wiring connector plug (see figure 2).

6.5.7 Enclosure. Enclosure consists of a sheet metal container which covers the sides, the top, and bottom of the switch. When switches are supplied with a wiring module, the enclosure includes a rear plate with the ship's wiring connector receptacles.

6.5.8 Description plate. A description plate is a plate that is marked to show panel assembly number, panel assembly nomenclature, and switch operating position information. Rotary switch position dial is also a description plate.

6.6 Special tools.

6.6.1 The following list of tools are ordered with and required in support of linear movement switches.

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List of required tools in support of switches.

Type of tools with MS number or manufacturer's part number					Intended use		
Crimp tool	FSCM	Turret head or die closure	Insertion tool	Extraction tool	Connector type or other	Contact type	Switch type
M22520/17-01	81349	N/A	M81969/13-01	M81969/11-01	M28731	M28731/20-0001 (16–26 AWG) M28731/23-0001 Double wire for: a. One 16 & one 20 AWG wire b. Two 18 AWG wires c. Two 20 AWG wires	All linear slide LS, BLS, CLS, DLS & KLS
P/N 06-7749-01	91662	N/A	M81969/13-01	M81969/11-01			
P/N 06-7748-01	91662	N/A	M81969/13-01	M81969/11-01		Two 16 AWG wires one 16 AWG and one 20 AWG	
M22520/17-01	81349	N/A	M81969/5-01	M81969/4-02		M28731/35	5BLS & 5DLS JLS
M22520/1-01	81349	M22520/1-02	M81969/17-01	M81969/19-01	M5015 MS3402 (P)	M29029/44-288 (16 AWG) -287 (22 AWG)	S3A3 & RC/SIP
M22520/1-01	81349	M22520/1-02	M81969/17-01	M81969/19-01	MS3406 (S)	M39029/45-295 (16 AWG) -294 (22 AWG)	
N/A		N/A	N/A	N/A		MS3187/16-2 Sealing plugs	
M22520/34-01 M22520/34-01	81349	M22520/34-02 M22520/34-02	M81969/33-01 M81969/33-01	M81969/34-01 M81969/34-01	M28840 (P) (Hi-density)	M39029/83-508 (20 AWG) -450 (22 AWG)	S3A3 & S3A4 RC/SIP for
M22520/34-01 M22520/34-01	81349	M22520/34-02 M22520/34-02	M81969/33-01 M81969/33-01	M81969/34-01 M81969/34-01	(S)	M39029/84-509 (20 AWG) -452 (22 AWG)	S3A4

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List of required tools in support of switches – Continued.

Type of tools with MS number or manufacturer's part number					Intended use		
Crimp tool	FSCM	Turret head or die closure	Insertion tool	Extraction tool	Connector type or other	Contact type	Switch type
N/A		N/A	N/A	N/A		MS27186-1 Sealing plugs	
M22520/1-01	81349	M22520/1-02	M81969/14-03	M81969/19-08	Relay socket (S) EL 410-14	M39029/5-116 (16 AWG)	S3A3 & S3A4
M22520/1-01	81349	M22520/1-02	M81969/14-03	M81969/19-08	Socket (S) MS12883/41-01	M39029/32-247 Relay	S3A3 & S3A4
M22520/1-01	81349	M22520/1-02	M81969/17-04	M81969/19-08	Relay (S) Socket	P/N 019-0215-002 16–#20 AWG	
M22520/5-01	81349	M22520/5-100			Terminal lug	MS17143	
					Terminal lug	MS35036	
					Terminal lug	MIL-T-7928/1	
					Terminal lug	MIL-T-7928/2	

Tools, other	FSCM	Type of tool	Intended use
M22520/3-1	81349	Inspection gauge	Used to inspect basic tool M22520/1-01
M22520/18-1	81349	Inspection gauge	Used to inspect tool M22520/17-01
M22520/3-9 & 10	81349	Inspection gauge	Used to inspect tool M22520/5-100 for 12 – 10 AWG for 26 – 14 AWG
P/N 11-6147-1 or equivalent	77820	Knurled ring pliers	Used to grip the knurled ring of circular type connectors: securing/removal
P/N 2020581	10001	Barrel nut driver (9 inches)	Used to install/withdraw barrel nuts on terminal boards
P/N 2020580	10001	Barrel nut driver (4 inches)	Used to install/withdraw barrel nuts on terminal boards

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6.7 Sub-contracted material and parts. The packaging or preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

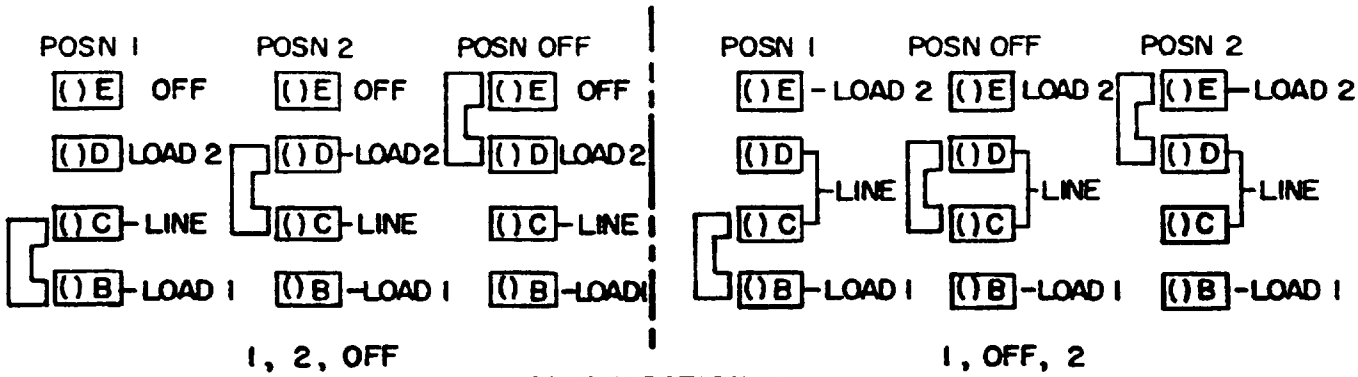
6.8 Subject term (key word) listing:

Circuit module
Connectors
Contact configurations
Fuses
Fuseholders
Lampholders
Load circuit operation
Module
Remote control module
Ships wiring connectors
Wire terminal

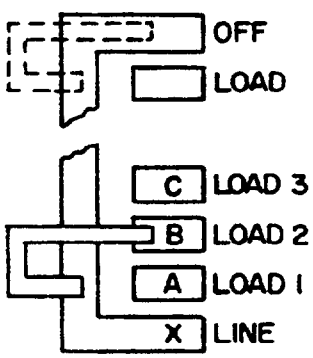
6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy – SH
(Project 5930-N681-01)

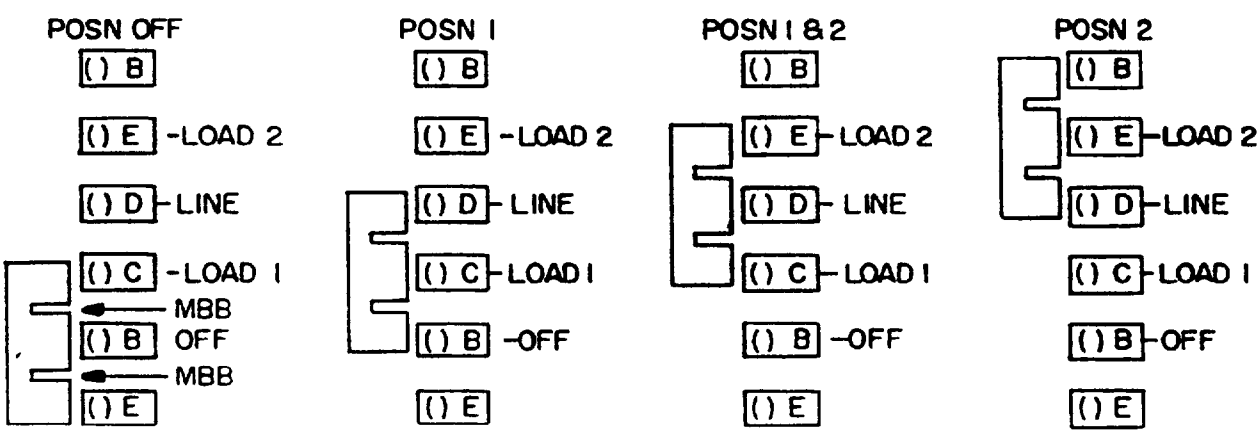
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CONFIGURATION 2



CONFIGURATION 3



CONFIGURATION 4

FIGURE 1 *Contact configurations*

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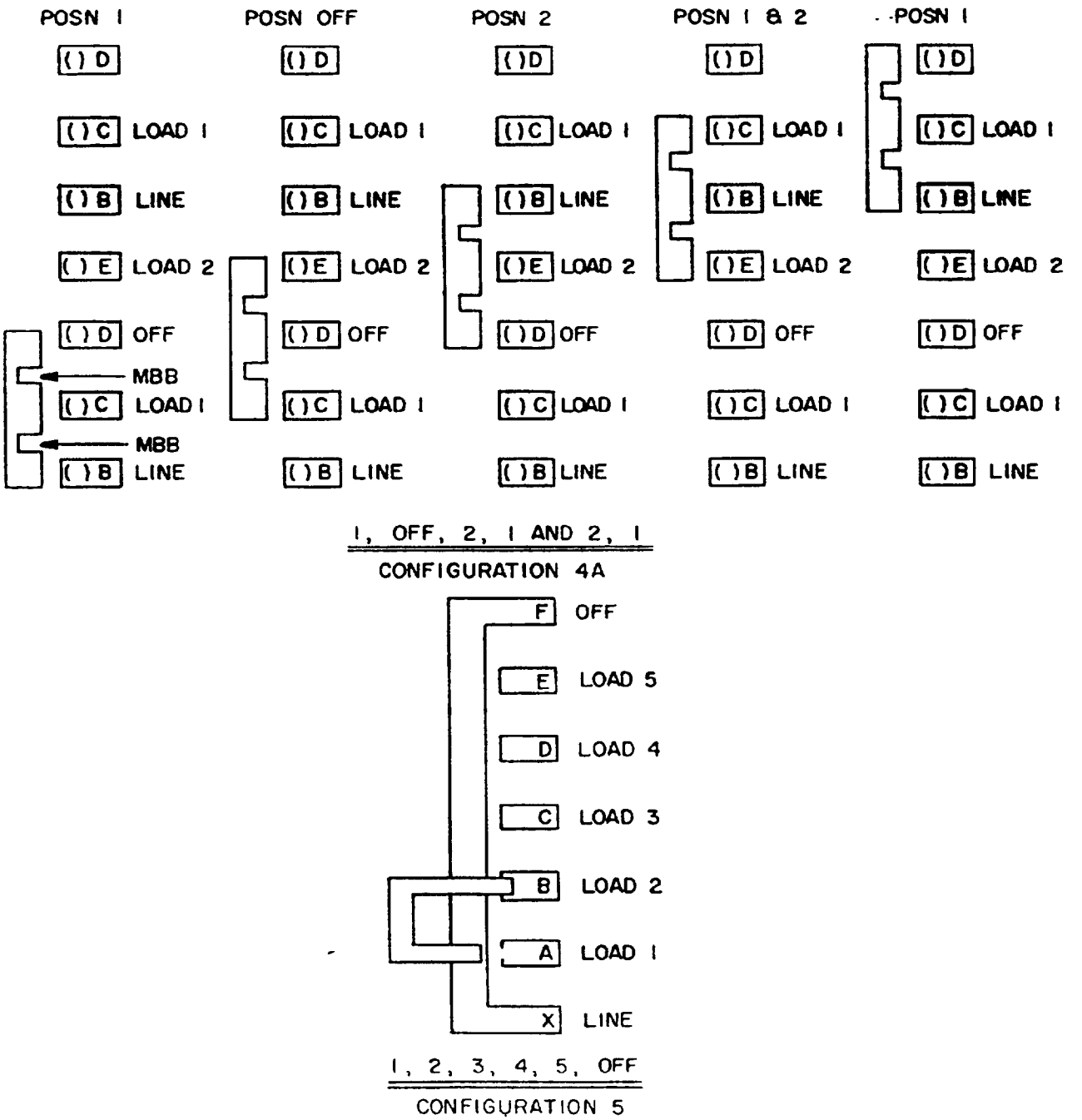


FIGURE 1 *Contact configurations – Continued*

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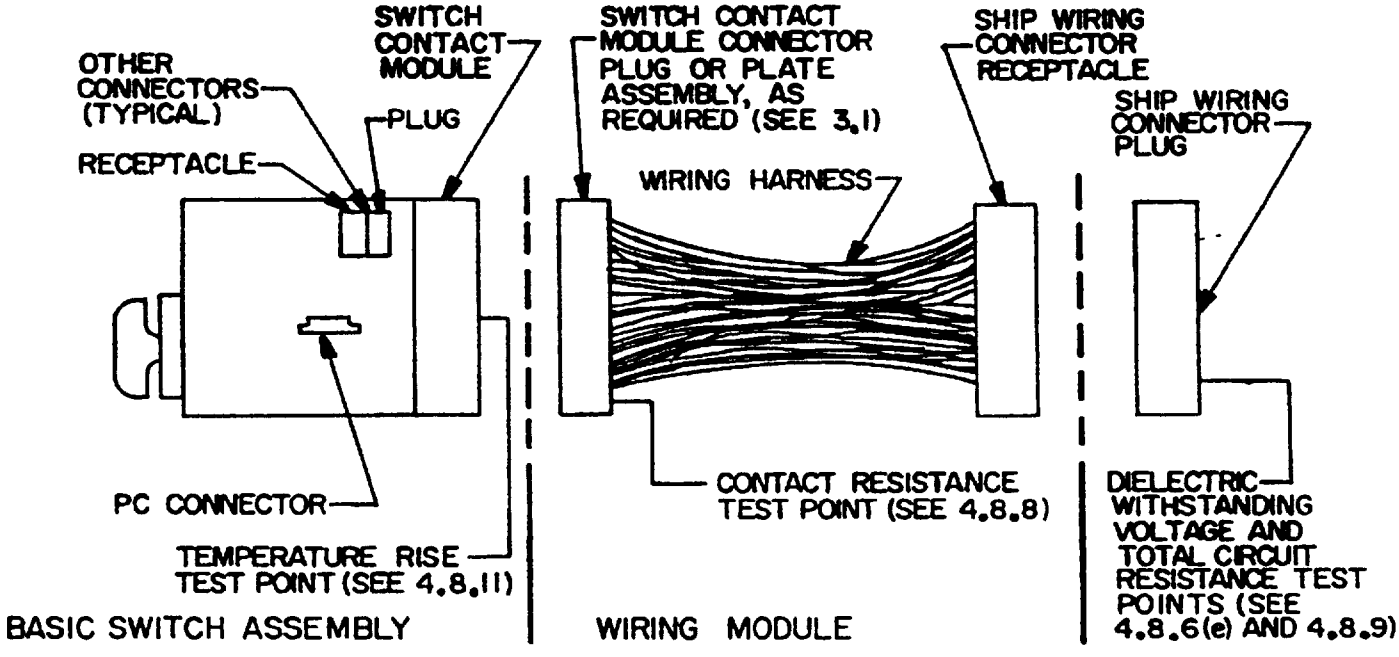


FIGURE 2. *Connector identification.*

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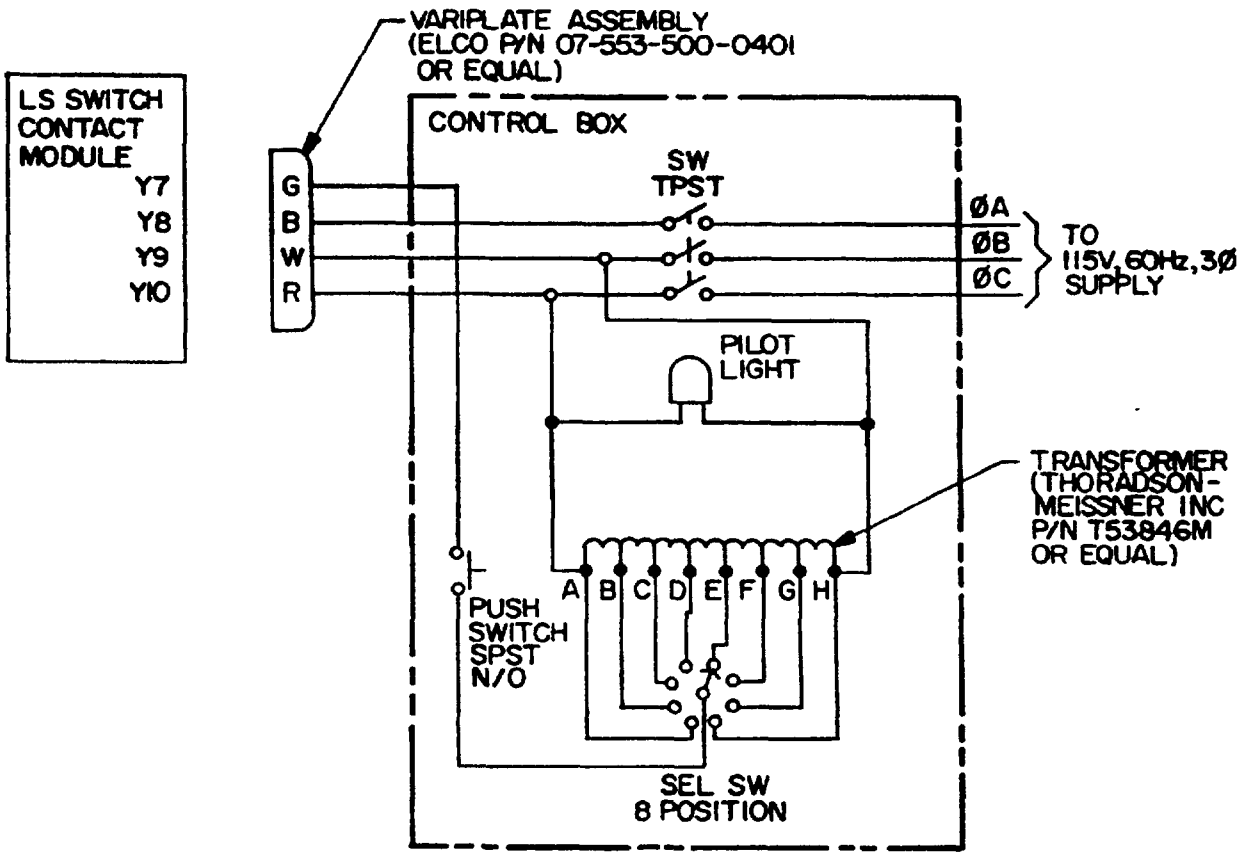


FIGURE 3. *Test circuit for remote operation of type RLS and RDLS switches.*

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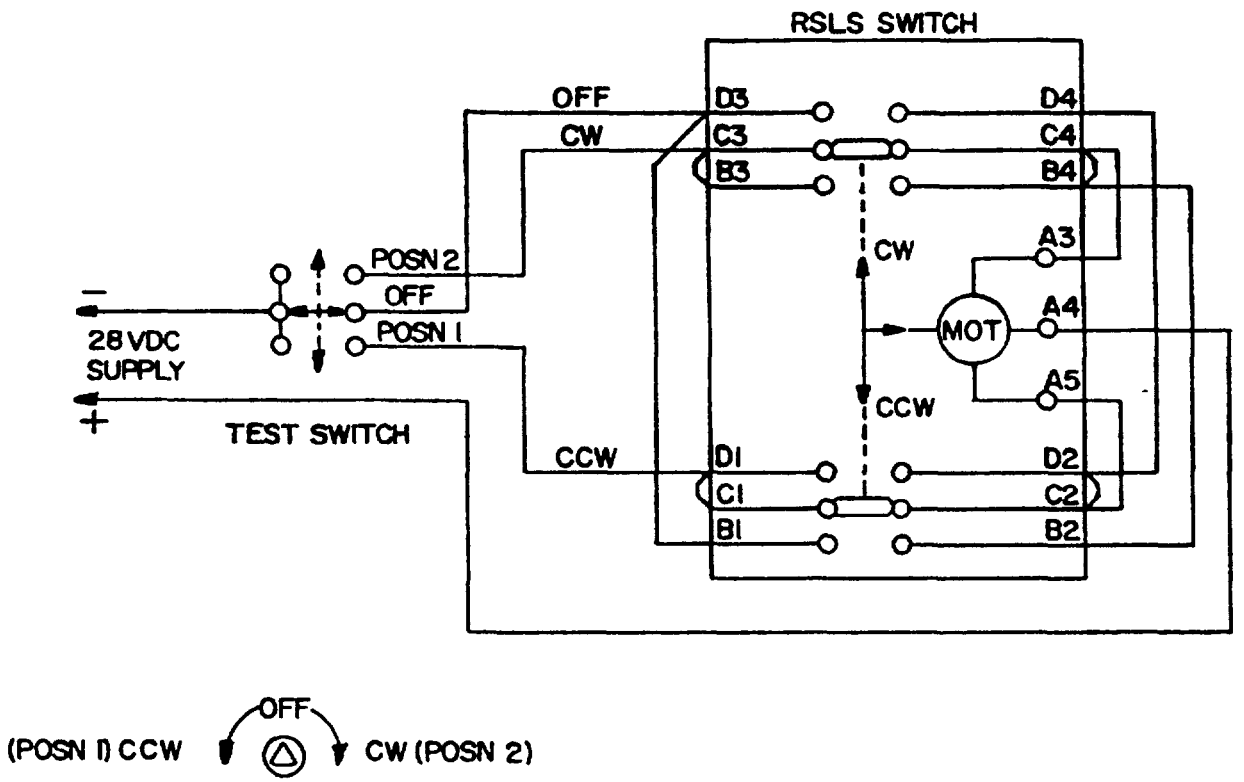


FIGURE 4. *Control circuit for test operation of RSLS switch.*

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APPENDIX

TECHNICAL INFORMATION REPORTS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers information that shall be included in the technical information reports when specified in the contract or order. This appendix is mandatory only when data item description DI-MISC-80652 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

20.1 Government document.

20.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

HANDBOOKS

MILITARY

MIL-HDBK-217 Reliability Prediction of Electronic Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

30. TECHNICAL INFORMATION

30.1 Reliability prediction. A reliability prediction shall be made on the equipment based upon electrical/mechanical/thermal stress analysis to assess the mean-time-between-failure (MTBF). The prediction shall be performed at the maximum ambient temperature condition and for the Naval sheltered environment. The prediction shall be in accordance with MIL-HDBK-217, and shall include the part reference designation, part type, part nomenclature, applicable Military part specification, part electrical/mechanical rating, design electrical/mechanical stress, electrical/mechanical stress ratio, manufacturer part thermal rating, design thermal stress, thermal stress ratio, base failure rate, failure rate application factors, final failure rate, and any other information used to determine part failure rates. Where MIL-HDBK-217 does not apply, other data sources may be used subject to review by the contracting activity When such alternate sources of data are proposed, sufficient information shall be provided to enable the contracting activity to evaluate its validity and applicability.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given
2. The submitter of this form must complete blocks 4, 5, 6, and 7
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NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-24187E(SH)	22 MARCH 1991

3. DOCUMENT TITLE

SWITCH ASSEMBLIES, LINEAR MOVEMENT, MANUAL AND REMOTE OPERATION, GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Technical Point of Contact (TPOC): Don Apperson (SEA 06D32) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS.	(1) Commercial TPOC: 703-602-2011	(2) AUTOVON 8-332-2011
c. ADDRESS (Include Zip Code) COMMANDER, NAVAL SEA SYSTEMS COMMAND DEPARTMENT OF THE NAVY (SEA 5523) WASHINGTON, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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